Complete SQL Code

Prepared by: Padmapriya Ganesh, Business Analyst – Jupiter

1. Create database, create table & alter table:

```
create database bankchurn;

create table bank_churn (
rownumber serial,
customerid varchar (20) primary key,
surname varchar (20),
creditscore int,
geography varchar (20),
gender varchar (10),
age int,
tenure int,
balance decimal,
numofproducts int,
hascrcard boolean,
isactivemember boolean,
estimatedsalary decimal,
exited boolean
);

alter table bank_churn
alter column surname type varchar
(50);
```

2. Importing CSV file into the table bank_churn

```
copy bank_churn(rownumber,
customerid, surname, creditscore,
geography, gender, age, tenure,
balance, numofproducts, hascrcard,
isactivemember, exited)
from
'C:\bankchurn\bank_churn_dataset.csv'
delimiter ','
csv header;
```

3. Computing Customer Churn %

```
select count (*) filter (where exited =
true) as total_customers_churned,
count (*) as total_customers,
ROUND((COUNT(*) FILTER (WHERE
exited = TRUE) * 100.0 / COUNT(*)), 2)
AS "churn percentage"
from bank_churn order by "churn
percentage" desc;
```

4. Exploring data to see distinct countries

```
select distinct geography from
bank_churn;
```

5. Geography-wise customer churn %

```
select geography,
    COUNT(*) FILTER (WHERE exited
= TRUE) AS exited_customers,
    COUNT(*) FILTER (WHERE exited
= FALSE) AS retained_customers,
          COUNT (*) AS
total_customers,
          ROUND(
     (COUNT(*) FILTER (WHERE
exited = TRUE) * 100.0 / COUNT(*)),
2) AS churn_percentage
from bank_churn
group by geography
order by churn_percentage desc;
```

6. Balance exit count (ranges with concat)

```
SELECT
  CONCAT(
    FLOOR(balance / 10000) *
10000,
    ' - ',
    FLOOR(balance / 10000) *
10000 + 9999
  ) AS balance_range,
  COUNT(*) AS exited_count
FROM
  bank_churn
WHERE
  exited = TRUE
  AND balance <= 300000
GROUP BY
  FLOOR(balance / 10000)
ORDER BY
  FLOOR(balance / 10000);
```

7. Balance Exited customers vs Retained customers

```
SELECT FLOOR(balance / 10000) *
10000 AS balance_range_start,
```

8. Balance Exited customers vs Retained customers with Churn %

```
SELECT CONCAT(
    FLOOR(balance / 10000) *
10000,
    ' - ',
    FLOOR(balance / 10000) *
10000 + 9999
  ) AS balance_range,
  COUNT(*) FILTER (WHERE exited =
TRUE) AS exited_count,
  COUNT(*) FILTER (WHERE exited =
FALSE) AS retained_count, count(*)
as total_customers,
ROUND((COUNT(*) FILTER (WHERE
exited = TRUE) * 100.0 / COUNT(*)),
2) AS churn_percentage
FROM bank_churn
WHERE balance <= 300000
GROUP BY FLOOR(balance / 10000)
ORDER BY churn_percentage desc;
```

9. Estimated Salary Exited customers vs Retained customers

```
SELECT FLOOR(estimatedsalary /
10000) * 10000 AS
salary_range_start,
FLOOR(estimatedsalary / 10000) *
10000 + 9999 AS salary_range_end,
COUNT(*) FILTER (WHERE exited =
TRUE) AS exited_count, COUNT(*)
FILTER (WHERE exited = FALSE) AS
stayed_count
FROM bank_churn WHERE
estimatedsalary <= 300000
GROUP BY FLOOR(estimatedsalary /
10000)
ORDER BY exited_count desc;
```

10. Estimated Salary Exited customers vs Retained customers with Churn %

```
SELECT CONCAT(
```

11. Age Exited customers vs Retained customers SELECT FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count FROM bank_churn WHERE age <= 100 GROUP BY floor(age/10) ORDER BY exited_count desc;	FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_true_count, COUNT(*) FILTER (WHERE exited = FALSE) AS exited_false_count FROM bank_churn WHERE balance <= 300000 GROUP BY FLOOR(balance / 10000) ORDER BY exited_true_count desc;	FLOOR(estimatedsalary / 10000) * 10000, ' - ', FLOOR(estimatedsalary / 10000) * 10000 + 9999) AS estimatedsalary_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage from bank_churn WHERE estimatedsalary <= 300000 GROUP BY FLOOR(estimatedsalary / 10000) ORDER BY churn_percentage desc;
12. Age Exited customers vs Retained customers with Churn % SELECT CONCAT(FLOOR(age/10) * 10, ' - ', FLOOR(age/10) * 10 + 9) AS age_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 GROUP BY floor(age/10) ORDER BY churn_percentage desc;	15. Geography + Age + Balance + Estimated salary wise churn % SELECT geography, FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end, FLOOR(balance / 10000) * 10000 AS balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 and balance <=300000 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(age/10), floor(balance/10000), geography ORDER BY churn_percentage desc;	16. Geography + Age + Balance + Estimated salary Exited customers vs Retained customers SELECT geography, FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end, FLOOR(balance / 10000) * 10000 AS balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers FROM bank_churn WHERE age <= 100 and balance <=300000 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(age/10), floor(balance/10000), geography ORDER BY exited_count desc;
13. Gender Exited customers vs Retained customers with Churn % select gender, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*)	17. Geography + Age + Estimated salary wise churn % SELECT geography, FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start,	18. Geography + Estimated salary Exited customers vs Retained customers SELECT geography, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) *

FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn GROUP BY gender ORDER BY churn_percentage desc;	FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(age/10), geography ORDER BY churn_percentage desc;	10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count FROM bank_churn WHERE estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), geography ORDER BY exited_count desc;
14. Geography-wise Exited customers vs Retained customers select geography, count (*) filter (where exited = true) as exited_count, count (*) filter (where exited = false) as stayed_count from bank_churn group by geography order by exited_count desc;	19. Geography + Salary wise churn % SELECT geography, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), geography ORDER BY churn_percentage desc;	20. Geography + Balance wise churn % SELECT geography, FLOOR(balance/ 10000) * 10000 AS balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE balance <=300000 GROUP BY FLOOR(balance / 10000), geography ORDER BY churn_percentage desc;
21. Combined Attribute Analysis **1. Geography + Age wise Churn %** SELECT geography, CONCAT(FLOOR(age/10) * 10, ' - ', FLOOR(age/10) * 10 + 9) AS age_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 GROUP BY floor(age/10), geography ORDER BY churn_percentage desc;	**22. Combined Attribute Analysis** **6. Gender + Age wise Churn %** SELECT gender, CONCAT(FLOOR(age/10) * 10, ' - ', FLOOR(age/10) * 10 + 9) AS age_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 GROUP BY floor(age/10), gender ORDER BY churn_percentage desc;	**23. Combined Attribute Analysis** **4. Geography + Gender wise Churn %** SELECT geography, gender, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn GROUP BY gender, geography ORDER BY churn_percentage desc;

24. Geography + Balance + Estimated salary wise churn %

```
SELECT geography,  FLOOR(balance / 10000) * 10000 AS balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn WHERE balance <=300000 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(balance/10000), geography
ORDER BY churn_percentage desc;
```

25. Combined Attribute Analysis
5. Gender + Credit score Churn %

```
SELECT gender,  CONCAT(
      FLOOR(creditscore / 100) * 100,
      ' - ',
      FLOOR(creditscore / 100) * 100 + 99
   ) AS creditscore_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn WHERE creditscore <=1000
GROUP BY FLOOR(creditscore / 100), gender ORDER BY churn_percentage desc;
```

26. Credit score Exited customers vs Retained customers with Churn %

```
SELECT CONCAT(
      FLOOR(creditscore / 100) * 100,
      ' - ',
      FLOOR(creditscore / 100) * 100 + 99
   ) AS creditscore_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn WHERE creditscore <=1000
GROUP BY FLOOR(creditscore / 100)
ORDER BY churn_percentage desc;
```

27. Combined Attribute Analysis
7. Geography + Credit score Churn %

```
SELECT geography, CONCAT(
      FLOOR(creditscore / 100) * 100,
      ' - ',
      FLOOR(creditscore / 100) * 100 + 99
   ) AS creditscore_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn WHERE creditscore <=1000 GROUP BY FLOOR(creditscore / 100), geography ORDER BY churn_percentage desc;
```

28. Credit Card Ownership Exited customers vs Retained customers with Churn %

```
SELECT hascrcard, COUNT(*) AS total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn
GROUP BY hascrcard
ORDER BY churn_percentage DESC;
```

29. Exploring the different number of products used by customers

```
select distinct numofproducts from bank_churn order by numofproducts;
```

30. Number of Products Exited customers vs Retained customers

```
select numofproducts,
   COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
   COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count
FROM
   bank_churn group by numofproducts order by exited_count desc;
select numofproducts, count (*) as customer_count from bank_churn group by numofproducts order by numofproducts asc;
```

33. Combined Attribute Analysis
3. Geography + Being an Active Member Churn %

```
SELECT geography, isactivemember,
   COUNT(*) AS total_customers,
   COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS
```

32. Combined Attribute Analysis
2. Geography + Number of products Churn %

```
select geography, numofproducts,
   COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
   COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
```

31. Number of Products Exited customers vs Retained customers with Churn %

```
select numofproducts,
   COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
   COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
```

retained_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn GROUP BY isactivemember, geography ORDER BY churn_percentage DESC;	count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn group by numofproducts, geography order by churn_percentage desc;	count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn group by numofproducts order by churn_percentage desc;
34. Being an Active Member Exited customers vs Retained customers with Churn % SELECT isactivemember, COUNT(*) AS total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn GROUP BY isactivemember ORDER BY churn_percentage DESC; 38. Country-wise Analysis: Churn Metrics for Spain (and Germany & France) (Same code – so didn't repeat by pasting them here in this document – for Germany & France for all 6 metrics, except for the final WHERE clause of the SQL SELECT queries: WHERE .... geography = 'Germany' & WHERE .... geography = 'France' respectively) 1. Salary SELECT geography, FLOOR(estimatedsalary / 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)" FROM	**35. Tenure Exited customers vs Retained customers with Churn %** select tenure, count (*) as total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage from bank_churn group by tenure order by churn_percentage desc; 37. Spain (Geography) + Age + Estimated salary wise churn % SELECT geography, FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)" FROM bank_churn WHERE geography = 'Spain' and age <= 100 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(age/10), geography ORDER BY "exited percentage ((exit/exit+stay)*100)" desc;	36. Spain (Geography) + Balance + Estimated salary wise churn % SELECT geography, FLOOR(balance / 10000) * 10000 AS balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, FLOOR(estimatedsalary/ 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)" FROM bank_churn WHERE geography = 'Spain' and balance <=300000 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(balance/10000), geography ORDER BY "exited percentage ((exit/exit+stay)*100)" desc;

```
   bank_churn
WHERE
   estimatedsalary <= 300000
   AND geography = 'Spain'
GROUP BY
   geography, FLOOR(estimatedsalary / 10000)
ORDER BY
   "exited percentage ((exit/exit+stay)*100)" DESC;
```

2. Balance

```
SELECT geography,
   FLOOR(balance / 10000) * 10000 AS balance_range_start,
   FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end,
   COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
   COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count,
   ROUND(
      (COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)),
      2
   ) AS "exited percentage ((exit/exit+stay)*100)"
FROM
   bank_churn
WHERE
   balance <= 300000
   AND geography = 'Spain'
GROUP BY
   geography, floor(balance/10000)
ORDER BY
   "exited percentage ((exit/exit+stay)*100)" DESC;
```

3. Credit Score

```
SELECT geography,
   CONCAT(
      FLOOR(creditscore / 100) * 100,
      ' - ',
      FLOOR(creditscore / 100) * 100 + 99
   ) AS creditscore_range,
   COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
   COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count,
   ROUND(
      (COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)),
      2
```

39. Summary stats for numerical variables

1. Estimated Salary

```
SELECT
   AVG(estimatedsalary) AS mean_salary,
   PERCENTILE_CONT(0.5)
      WITHIN GROUP (ORDER BY estimatedsalary) AS median_salary,
   MODE()
      WITHIN GROUP (ORDER BY estimatedsalary) AS mode_salary,
   MIN(estimatedsalary) AS min_salary,
   MAX(estimatedsalary) AS max_salary,
   STDDEV(estimatedsalary) AS stddev_salary
FROM
   bank_churn;
```

2. Balance

```
SELECT
   AVG(balance) AS mean_balance,
   PERCENTILE_CONT(0.5)
      WITHIN GROUP (ORDER BY balance) AS median_balance,
   MODE()
      WITHIN GROUP (ORDER BY balance) AS mode_balance,
   MIN(balance) AS min_balance,
   MAX(balance) AS max_balance,
   STDDEV(balance) AS stddev_balance
FROM
   bank_churn;
```

3. Credit Score

```
SELECT
   AVG(creditscore) AS mean_creditscore,
   PERCENTILE_CONT(0.5)
      WITHIN GROUP (ORDER BY creditscore) AS median_creditscore,
   MODE()
      WITHIN GROUP (ORDER BY creditscore) AS mode_creditscore,
   MIN(creditscore) AS min_creditscore,
   MAX(creditscore) AS max_creditscore,
   STDDEV(creditscore) AS stddev_creditscore
FROM
   bank_churn;
```

40. Spain geography, creditscore, balance, number of products, estimated salary, age & gender, ordered by highest balance

```
select geography, creditscore, balance, numofproducts, estimatedsalary, age, gender from bank_churn where geography = 'Spain' and exited = true order by balance desc;
```

```
    ) AS "exited percentage
((exit/exit+stay)*100)"
FROM
    bank_churn
WHERE
    creditscore <= 1000
    AND geography = 'Spain'
GROUP BY
    geography, floor(creditscore/100)
ORDER BY
    "exited percentage
((exit/exit+stay)*100)" DESC;
```

4. Age

```
SELECT geography,
    FLOOR(age / 10) * 10 AS
age_range_start,
    FLOOR(age / 10) * 10 + 9 AS
age_range_end,
    COUNT(*) FILTER (WHERE exited =
TRUE) AS exited_count,
    COUNT(*) FILTER (WHERE exited =
FALSE) AS stayed_count,
    ROUND(
        (COUNT(*) FILTER (WHERE exited =
TRUE) * 100.0 / COUNT(*)),
        2
    ) AS "exited percentage
((exit/exit+stay)*100)"
FROM
    bank_churn
WHERE
    age <= 100
    AND geography = 'Spain'
GROUP BY
    geography, floor(age/10)
ORDER BY
    "exited percentage
((exit/exit+stay)*100)" DESC;
```

5. Gender

```
SELECT geography, gender,
    COUNT(*) FILTER (WHERE exited =
TRUE) AS exited_count,
    COUNT(*) FILTER (WHERE exited =
FALSE) AS stayed_count,
    ROUND(
        (COUNT(*) FILTER (WHERE exited =
TRUE) * 100.0 / COUNT(*)), 2) AS
"exited percentage
((exit/exit+stay)*100)"
FROM
    bank_churn
WHERE geography = 'Spain'
GROUP BY
```

4. Age

```
SELECT
    AVG(age) AS mean_age,
    PERCENTILE_CONT(0.5)
        WITHIN GROUP (ORDER BY age)
AS median_age,
    MODE()
        WITHIN GROUP (ORDER BY age)
AS mode_age,
    MIN(age) AS min_age,
    MAX(age) AS max_age,
    STDDEV(age) AS stddev_age
FROM
    bank_churn;
```

5. Tenure

```
SELECT
    AVG(tenure) AS mean_tenure,
    PERCENTILE_CONT(0.5)
        WITHIN GROUP (ORDER BY
tenure) AS median_tenure,
    MODE()
        WITHIN GROUP (ORDER BY
tenure) AS mode_tenure,
    MIN(tenure) AS min_tenure,
    MAX(tenure) AS max_tenure,
    STDDEV(tenure) AS stddev_tenure
FROM
    bank_churn;
```

6. Number of Products

```
SELECT
    AVG(numofproducts) AS
mean_numofproducts,
    PERCENTILE_CONT(0.5)
        WITHIN GROUP (ORDER BY
numofproducts) AS
median_numofproducts,
    MODE()
        WITHIN GROUP (ORDER BY
numofproducts) AS
mode_numofproducts,
    MIN(numofproducts) AS
min_numofproducts,
    MAX(numofproducts) AS
max_numofproducts,
    STDDEV(numofproducts) AS
stddev_numofproducts
FROM
    bank_churn;
```

geography, gender ORDER BY "exited percentage ((exit/exit+stay)*100)" DESC; 6. Number of Products SELECT geography, numofproducts, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)" FROM bank_churn WHERE geography = 'Spain' GROUP BY geography, numofproducts ORDER BY "exited percentage ((exit/exit+stay)*100)" DESC;		

FINAL QUERY

```
WITH customer_risk AS (
  SELECT customerid, geography, age, creditscore, balance, estimatedsalary, numofproducts, isactivemember,
    hascrcard, tenure, gender, exited,
    -- risk scoring logic based on findings from report
    (CASE
      WHEN isactivemember = FALSE THEN 2 ELSE 0 END +
     CASE
      WHEN numofproducts = 1 THEN 2
      WHEN numofproducts >= 3 THEN 3
      ELSE 0 END +
     CASE
      WHEN geography = 'Germany' THEN 2 ELSE 0 END +
     CASE
      WHEN age BETWEEN 40 AND 69 THEN 2 ELSE 0 END
    ) AS churn_risk_score,
```

```
    -- window ranking
    ROW_NUMBER() OVER (
      PARTITION BY exited
      ORDER BY
        (CASE
          WHEN isactivemember = FALSE THEN 2 ELSE 0 END +
        CASE
          WHEN numofproducts = 1 THEN 2
          WHEN numofproducts >= 3 THEN 3
          ELSE 0 END +
        CASE
          WHEN geography = 'Germany' THEN 2 ELSE 0 END +
        CASE
          WHEN age BETWEEN 40 AND 69 THEN 2 ELSE 0 END
        ) DESC
    ) AS risk_rank
  FROM bank_churn
)
SELECT * FROM customer_risk WHERE risk_rank <= 10 ORDER BY exited, risk_rank;
```

OUTPUT:

	customerid [PK] character varying (20)	geography character varying (20)	age integer	creditscore integer	balance numeric	estimatedsalary numeric	numofproducts integer	isactivemember boolean	hascrcard boolean	tenure integer	gender character	exited boolean	churn_risk_score integer	risk_rank bigint
1	15794941	Germany	41	647	85906.65	189159.97	3	false	true	1	Female	false	9	1
2	15771728	Germany	41	641	104405.54	17384.21	3	false	true	7	Male	false	9	2
3	15677307	Germany	40	684	137326.65	186976.6	1	false	true	6	Female	false	8	3
4	15689294	Germany	44	705	105934.96	82463.69	1	false	true	3	Male	false	8	4
5	15614813	Germany	46	777	107362.8	487.3	1	false	true	0	Female	false	8	5
6	15738884	Germany	41	642	157777.58	67484.6	1	false	true	4	Male	false	8	6
7	15724466	Germany	41	744	84113.41	197548.63	1	false	true	2	Female	false	8	7
8	15770995	Germany	47	753	131160.85	197444.69	1	false	true	1	Female	false	8	8
9	15636820	Germany	40	725	104149.66	62027.9	1	false	true	8	Male	false	8	9
10	15744914	Germany	42	539	177728.55	105013.63	1	false	true	1	Male	false	8	10
11	15710087	Germany	54	705	125889.3	96013.5	3	false	true	3	Female	true	9	1
12	15589017	Germany	55	547	111362.76	16922.28	3	false	true	4	Male	true	9	2
13	15739972	Germany	45	650	152367.21	150835.21	3	false	true	9	Female	true	9	3
14	15727619	Germany	46	753	113909.69	92320.37	3	false	true	9	Female	true	9	4
15	15711843	Germany	40	613	147856.82	107961.11	3	false	false	1	Male	true	9	5
16	15701602	Germany	52	521	116497.31	53793.1	3	false	false	5	Male	true	9	6
17	15640846	Germany	58	546	106458.31	128881.87	4	false	true	3	Female	true	9	7
18	15775761	Germany	69	610	86038.21	192743.06	3	false	false	5	Female	true	9	8
19	15580148	Germany	40	750	168286.81	20451.99	3	false	true	5	Male	true	9	9
20	15694376	Germany	64	705	153469.26	146573.66	3	false	false	3	Female	true	9	10

.